MILITARY SPECIFICATION

TRAILER, RAIL TYPE, LIFT ETU-77E, 3,000-POUND CAPACITY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 3,000-pound capacity, rail type, lift trailer, designated ETU-77/E.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60	Tape Packaging, Waterproof

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-4727	Trailers, Rail-Type, General Specification for.
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-H-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-C-52950	Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments).
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

DRAWING

PRODUCT DEVELOPMENT GROUP, INCORPORATED

PDG-0327B	Trailer, Lift, Rail Type, 3,000 Pound Capacity AF Type ETU-77( )/E.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82	Standard Practice For Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article testing.

3.2 General requirements. The requirements of MIL-T-4727 as specified for type I trailers and the requirements of Drawing PDG-0327B and details thereto apply as requirements of this specification with the exceptions and additions specified herein (see 6.2). When MIL-T-4727 and the drawings conflict with this specification, this specification shall govern.

* 3.2.1 Exceptions to the requirements of MIL-T-4727. The requirements for the winch, mobility, air transportability, and the environmental conditions are not applicable to this specification.

3.3 Design and contruction.

3.3.1 Maintainability. The maintainability requirements shall be in accordance with MIL-STD-470.

3.3.2 Life expectancy. The trailer shall be designed for a life expectancy of 10 years.

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3.3.3 Capacity. The trailer shall have a rated capacity of 3,000 pounds.

3.4 Performance.

3.4.1 Proofload. The trailer shall withstand a proofload of 200 percent of rated capacity. The center of gravity shall be 12 inches above the center of the parallel rails and applied to two points on each rail 40 inches apart.

3.4.2 Hydraulic lift system. The hydraulic lift system shall be redesigned as necessary to withstand the lifting cycles specified herein without leakage or failure of the system. Each cylinder shall be equipped with a position-locking, fail-safe device to prevent lowering or tipping of the rails in the event of hydraulic failure.

3.4.3 Lift system controls. The hydraulic lift system and controls shall be redesigned as necessary and shall be located in one operator's position, enabling the operator to raise the rails maintaining a level attitude. At this location, the hydraulic system shall be equipped with quick-disconnect line couplings in order to provide a power driven hydraulic system to the unit. The reservoir shall have ample capacity to accommodate any additional fluid from the power source.

3.5 Weight. The weight of the trailer shall not exceed 1,600 pounds.

3.6 Finish. The trailer shall be finished in yellow, color designation DG.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the trailer shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance tests (see 4.5).

4.3 Test conditions. During testing of the trailer, the hydraulic system shall be serviced with hydraulic fluid conforming to MIL-H-5606.

4.4 First article testing (see 6.2)

4.4.1 Test sample. One trailer shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

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4.4.3 First article tests. The first article rests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following tests as described under 4.6:

a. Examination of product (see 4.6.1)

b. Load test (see 4.6.2)

4.6 Test methods.

4.6.1 Examination of product. The trailer shall be visually examined to determine compliance with the requirements specified herein with respect to materials, quality of workmanship, design and finishes, markings, defects, and the functional operation of all components, assemblies, accessories and controls, and compliance with the maintainability requirements. Particular attention shall be given to leakage in the hydraulic system, and to routing and protection of hydraulic lines.

4.6.2 Load test. The trailer shall be tested by placing the rated load in the center of the rails supported from two points on each rail and raised. The load shall then be shifted to each end of the trailer and the suitability of the hydraulic cylinders shall be proven. The load shall be so applied as to enable compression and tension loads to be applied to the lift cylinders. During this test, all manipulators shall be operated through their entire ranges. No binding or roughness of operation shall be evident.

4.6.3 Endurance test. An endurance test shall be conducted on the hydraulic system of the trailer. The test shall consist of 50 cycles of raising the rated load, one cycle consisting of raising the trailer from collapsed height to the fully extended height and return to the collapsed position. At the end of the endurance test, the force required at the end of the pump handle(s) to raise the load shall not exceed the loads specified on the drawings. After the endurance test, the cylinder settling rate shall not exceed 0.020 inch per height adjustment.

4.6.4 Static load tests. The trailer shall be subjected to a static load of 200 percent of the rated load. The load shall be raised to maximum height and lowered to the low position while loaded. The lifts provided on the wheels shall also be subjected to this static load and operated through their entire ranges. No damage shall result from the static load. The lift cylinder shall exhibit no leaks around the seals and the hydraulic system shall be suitable for these entire loads.

4.6.4.1 Settling tests. During the test specified in 4.6.4, the settling rate of the hydraulic cylinders shall be measured. The cylinder settling rate shall not exceed 0.02 inch per hour per cylinder. The hydraulic locks shall not be used during this test.

* 4.6.4.2 Adjustment limit tests. The tilt, roll, yaw, and transverse mechanism shall be operated through their entire ranges and shall be tested under 200 percent rated load. The adjusting mechanism shall be operable by a drive wrench as specified on the drawings. If the torque required to perform all adjustment exceeds the amount specified on the drawings, the trailer shall be rejected.

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4.6.4.3 Side load test. The trailer shall be raised to its maximum height. The trailer shall be subjected to a static load of 200 percent rated load. A side load of 10 percent rated load shall be applied at the center-of-gravity of the load a maximum distance of 12 inches above the rails. The trailer shall not show permanent deformation, tip, or overturn when the side load is applied. The trailer shall not be tied down during the test.

* 4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Standard Practice for Commercial Packaging, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each trailer shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1, D-4, or D-5 of MIL-P-116 as applicable.

5.1.1.2 Unit protection. Each trailer shall be preserved and packaged in accordance with method I of MIL-P-116. The hydraulic system shall be filled with hydraulic fluid conforming to MIL-H-6083; openings shall be covered with greaseproof paper conforming to MIL-B-121 and sealed with water-resistant tape conforming to PPP-T-60. A waterproof tag clearly marked as follows shall be secured to the trailer:

CAUTION

Hydraulic system filled with preservative hydraulic fluid. Drain and replace with operational hydraulic fluid before placing equipment in service.

* 5.1.2 Commercial. Commercial packaging shall be in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B or commercial, as specified (see 6.2).

* 5.2.1 Level A. The trailer, preserved as specified in 5.1.1, shall be adjusted to its lowest position and packed in a crate conforming to MIL-C-52950, style B type V, covered. The tow bar shall be removed prior to crating and secured to the unit. Unless otherwise specified by the procuring activity, this level shall apply to overseas shipment only.

5.2.2 Level B. The trailer, preserved and packaged as specified in 5.1.1, shall be shipped as a mobile unit.

* 5.2.3 Commercial. The trailer shall be packed in accordance with the requirements of ASTM D 3951.

* 5.3 Physical protection. Cushioning, anchoring, bracing, and blocking shall be in accordance with MIL-STD-1186.

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* 5.4 Marking. In addition to any special markings required by the contract or order, interior and exterior containers and trailers shipped devoid of container shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable. The shipment marking nomenclature shall be as follows:

Trailer, Rail Type, Lift ETU--7/E, 3,000 Pound Capacity

6. NOTES

6.1 Intended use. The ETU-77/E trailer is intended for use in installing and removing engines and other components from aircraft and missiles, and for the ground transportation and handling of these components.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawings (see 3.2) in order to achieve the performance requirements of this specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of this contract.

c. Location and conditions for first article testing (see 4.4).

d. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

* 6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-38403A	LIFE PRESERVER, UNDERARM, CASUALTY, TYPE MB-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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